

求零点/极点

1. 2法则(估算)

适用: 简单的单极共源、共栅、共源共栅放大器

节点电容很多, 只要求极点

求法: ① 选择一个电容, 将其他电容开路

② 计算从C看进去的等效电容R

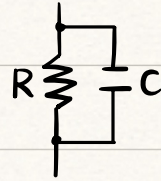
$$\textcircled{3} \tau = RC, f = \frac{1}{2\pi RC}$$

2. 辅助定理求解

适用: 电容和电阻并联

求法: ① 列出不考虑电容的表达式

$$\textcircled{2} R \rightarrow R \parallel \frac{1}{sC}$$



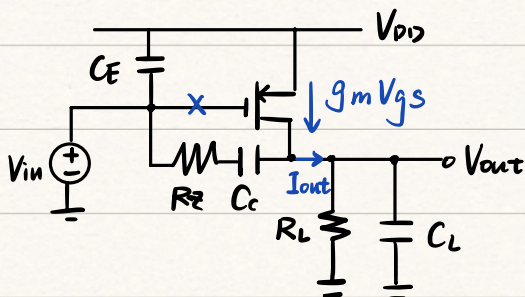
3. 小信号图求解

求法: ① 画小信号图

② 电容看作 $\frac{1}{sC}$ 的电阻 $\rightarrow$ 求 A_v $\rightarrow$ 零点

③ 整理表达式 $A_v = A_{vm} \cdot \frac{1 + sC_3 R_3}{(1 + sC_1 R_1)(1 + sC_2 R_2)} \rightarrow$ 极点

拉扎维书上求零点:



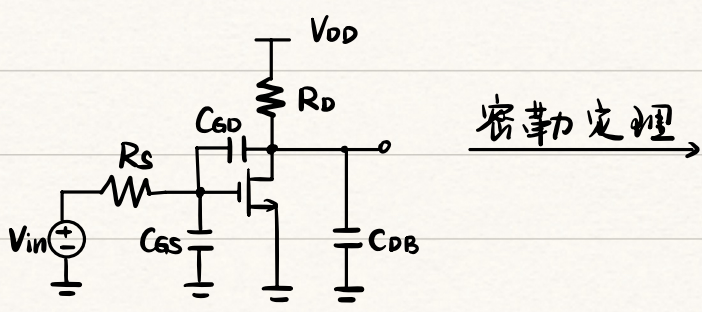
$$I_{out} = g_m V_{gs} + \frac{V_{gs}}{R_D + \frac{1}{sC_C}} = 0$$

$\downarrow$

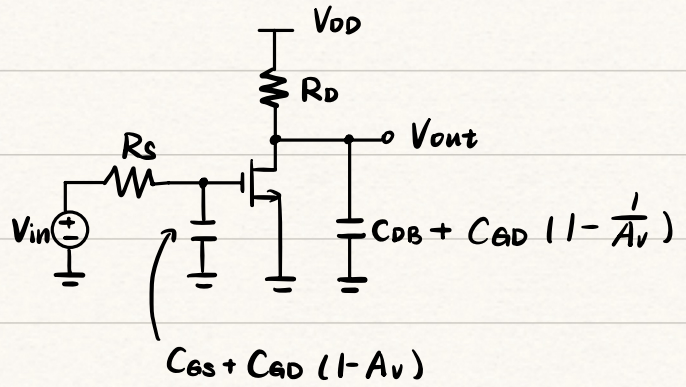
$$1 + sC_C (R_D - \frac{1}{g_m}) = 0$$

$\downarrow$ ω_z

共源极



密勒定理

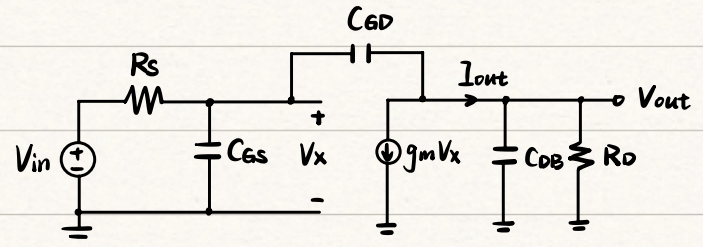


极点 $\left\{ \begin{aligned} \omega_{in} &= \frac{1}{R_s [C_{gs} + (1 + g_m R_D) C_{gd}]} \end{aligned} \right.$

$\omega_{out} = \frac{1}{R_D [C_{db} + (1 + \frac{1}{g_m R_D}) C_{gd}]} \approx \frac{1}{R_D (C_{db} + C_{gd})}$

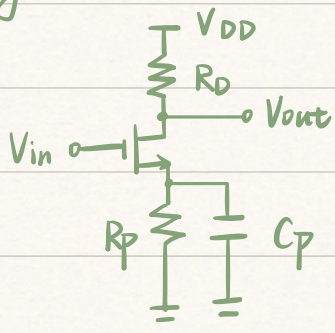
零点: $\frac{V_x - V_{out}}{1/sC_{gd}} = g_m V_x$

$\xrightarrow{V_{out}=0} 1 - \frac{s}{g_m/C_{gd}} = 0 \Rightarrow \omega = \frac{g_m}{C_{gd}}$
恒定零点



传输函数: $\frac{V_{out}}{V_{in}} = \frac{-g_m R_D}{(1 + \frac{s}{\omega_{in}})(1 + \frac{s}{\omega_{out}})} \cdot (1 - \frac{s}{g_m/C_{gd}})$

eg.1 求 Av



共源: $A_{v本} = -g_m r_o$

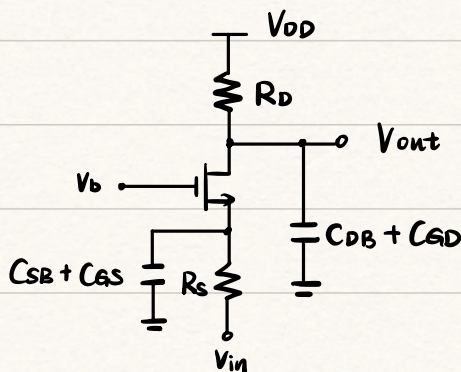
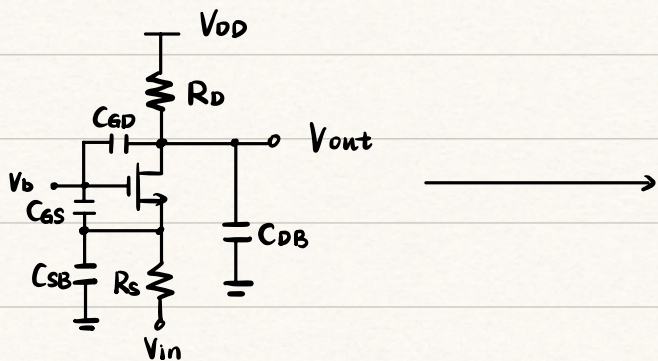
$R_L = R_D$

$R_{out} = r_o + [1 + (g_m + g_{mb}) r_o] (R_p \parallel \frac{1}{sC_p})$

$A_v = -g_m r_o \cdot \frac{R_D}{R_D + r_o + [1 + (g_m + g_{mb}) r_o] (R_p \parallel \frac{1}{sC_p})}$

传输函数: $A_v = \frac{-g_m r_o R_D}{R_D + r_o + [1 + (g_m + g_{mb}) r_o] R_p} \cdot \frac{(1 + sC_p R_p)}{1 + \frac{(R_D + r_o) sC_p R_p}{R_D + r_o + [1 + (g_m + g_{mb}) r_o] R_p}}$

共栅极 (无零点)

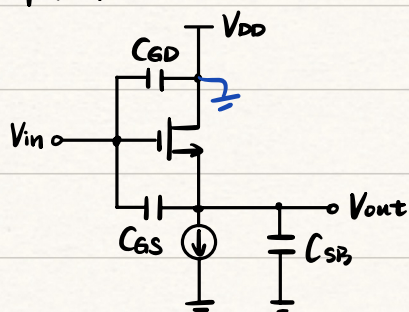


$$W_{in} = (R_S \parallel \frac{1}{g_m + g_{mb}}) (C_{GS} + C_{SB})$$

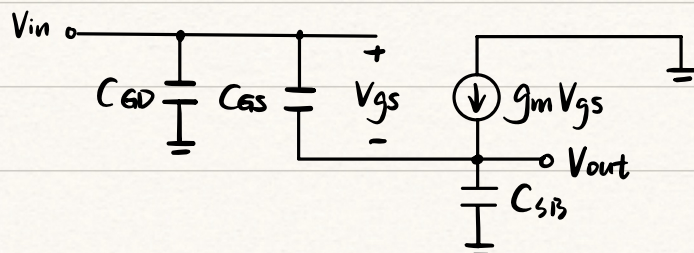
$$W_{out} = \frac{1}{R_D (C_{DB} + C_{GD})}$$

传输函数: $A_v = \frac{(g_m + g_{mb}) R_D}{1 + (g_m + g_{mb}) R_S} \cdot \frac{1}{(1 + \frac{S}{W_{in}})(1 + \frac{S}{W_{out}})}$

共漏极



小信号图



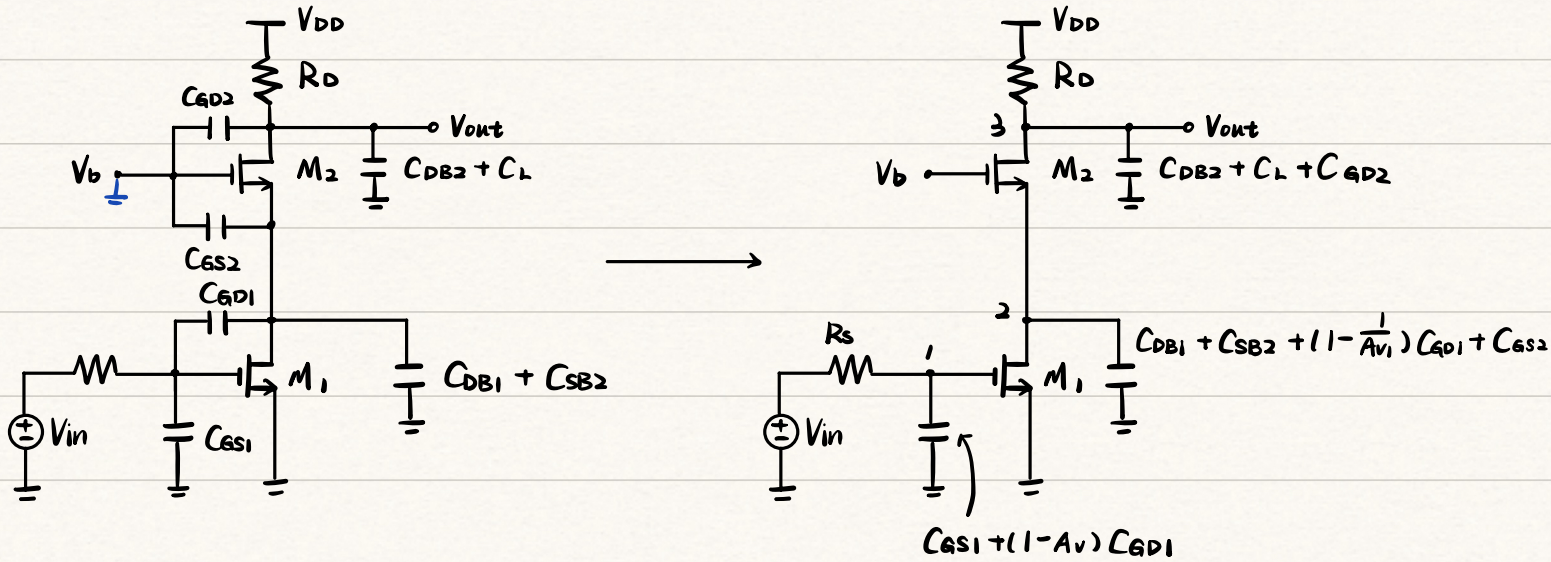
$$\left\{ \begin{aligned} \frac{V_{in} - V_{out}}{\frac{1}{SC_{GS}}} + g_m V_{gs} &= \frac{V_{out}}{\frac{1}{SC_{SB}}} \\ \Rightarrow \frac{V_{out}}{V_{in}} &= \frac{g_m + C_{GS} \cdot S}{g_m + (C_{SB} + C_{GS}) \cdot S} \end{aligned} \right.$$

$$V_{gs} = V_{in} - V_{out}$$

传输函数: $\frac{V_{out}}{V_{in}} = \frac{1 + \frac{S}{g_m/C_{GS}}}{1 + \frac{S}{g_m/(C_{SB} + C_{GS})}}$

$$\left\{ \begin{aligned} W_Z &= g_m / C_{GS} \\ W_P &= g_m / (C_{SB} + C_{GS}) \end{aligned} \right.$$

共源共栅极 (无零点)



$$A_{v1} = -g_{m1} \cdot \frac{1}{g_{m2} + g_{mb2}}$$

$$C_1 = C_{gs1} + \left(1 + \frac{g_{m1}}{g_{m2} + g_{mb2}}\right) C_{gd1}, \quad R_1 = R_s$$

$$C_2 = C_{db1} + C_{sb2} + C_{gs2} + \left(1 + \frac{g_{m2} + g_{mb2}}{g_{m1}}\right) C_{gd1}, \quad R_2 = \frac{1}{g_{m2} + g_{mb2}}$$

$$C_3 = C_{db2} + C_{gd2} + C_L, \quad R_3 = R_D$$

频率补偿

1. 目的：避免产生自激振荡

2. 方法

① 滞后补偿：引入低频半的主极点，使增益快速下降

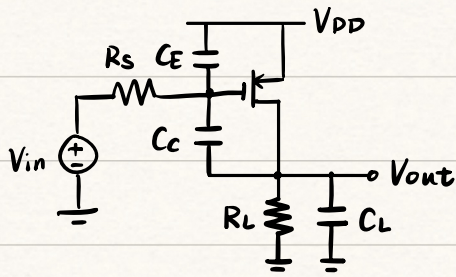
常用：密勒电容 C_c

缺点：降低带宽

② 超前补偿：引入零点，产生超前相位

常用：在 C_c 上串联一个 $R_c \rightarrow f_z = \frac{1}{2\pi R_c C_c}$

eg. 密勒电容



引入 C_c :

缺点: 引入不好的零点

优点: 栅极极点 ↓, 输出极点 ↑

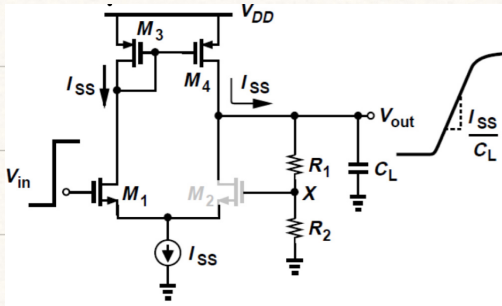
↓
极点分离有利于稳定
主极点

压摆率 SR

1. 定义: 运算放大器在大信号输入条件, 输出电压所能达到的最大变化速率

来自于内部电路 I_{out} 的限制

$$2. SR = \frac{I_{ss}}{C_L}$$



§4. 噪声

特性

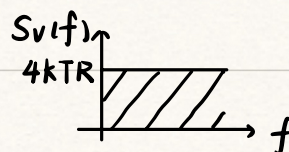
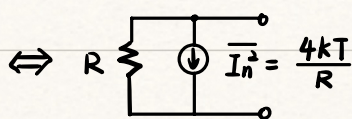
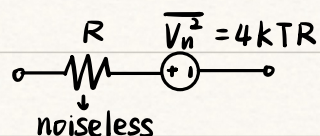
瞬态不可预测但平均功率固定 $\Rightarrow$ 求的都是平均值

将输出噪声等效到输入端: $\overline{V_{n.in}^2} = \frac{\overline{V_{n.out}^2}}{A_v^2}$, $\overline{V_{out}^2} = \overline{I_{out}^2} \cdot R_{out}^2$

热噪声 (白噪声)

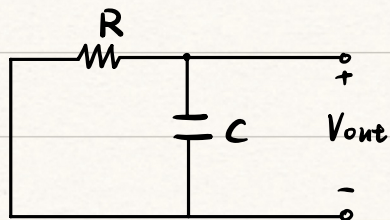
1. 电阻

等效模型:

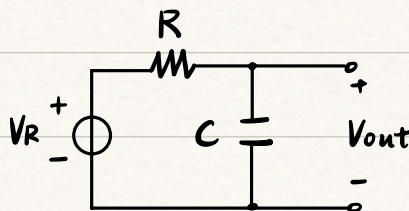


一般将噪声放在靠近电源或地端

2. RC 网络



考虑噪声 $\rightarrow$



$$\frac{V_{out}}{V_R} = \frac{1}{1 + sCR}$$

$$S_{out}(f) = S_R(f) \cdot A_v^2 = 4kTR \cdot \frac{1}{1 + 4\pi^2 R^2 C^2 f^2} \quad \left\{ \begin{array}{l} s = j\omega = j \cdot 2\pi f \\ \text{求的是 } A_v \text{ 模长的平方} \end{array} \right.$$

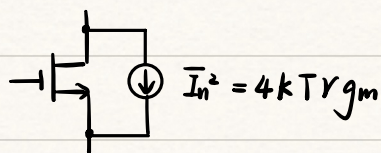
$$P_{out}(f) = \int_0^{+\infty} S_{out}(f) df = 4kTR \cdot \frac{1}{2\pi RC} \arctan(2\pi RCf) \Big|_0^{+\infty} = \frac{kT}{C}$$

电阻引入噪声, 噪声功率仅与电容大小有关

Mos 器件噪声

1. 热噪声

等效模型:

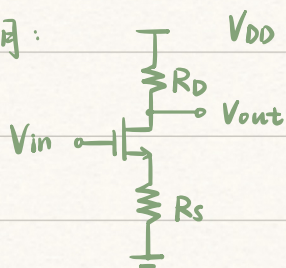


γ : 工艺相关系数, 长沟道中一般取 $\frac{2}{3}$

$$\bar{I}_n^2 = g_m^2 \bar{V}_{gs}^2$$

栅极的电压噪声永远是 $4kT\gamma \cdot \frac{1}{g_m}$

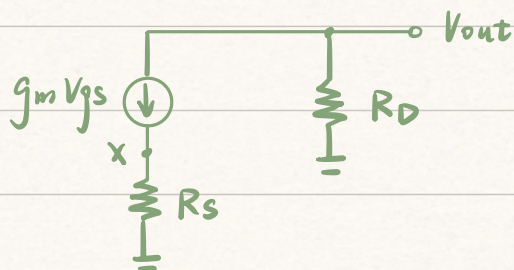
证明:



$$R_{out} = \frac{R_D \cdot [r_o + (1 + g_m r_o) R_S]}{R_D + r_o + R_S + g_m r_o R_S}$$

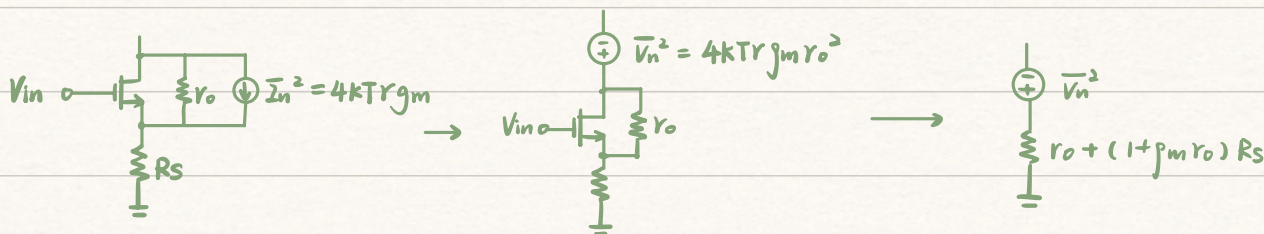
$\downarrow r_o \rightarrow \infty$

$$= \frac{(1 + g_m R_S) R_D}{1 + g_m R_S} = R_D$$



$$\begin{cases} V_{out} = -g_m (V_{in} - V_x) R_D \\ g_m (V_{in} - V_x) = \frac{V_x}{R_S} \end{cases}$$

$$A_v = - \frac{g_m R_D}{1 + g_m R_S}$$



$$\bar{I}_{out}^2 = \frac{4kT\gamma g_m r_o^2}{[r_o + (1 + g_m r_o) R_S]^2} \xrightarrow{r_o \rightarrow \infty} \bar{I}_{out}^2 = \frac{4kT\gamma g_m}{(1 + g_m R_S)^2}$$

$$\bar{V}_{out}^2 = \bar{I}_{out}^2 \cdot R_{out}^2 = 4kT\gamma g_m R_D^2 \cdot \left(\frac{1}{1 + g_m R_S} \right)^2$$

$$\bar{V}_{in}^2 = \frac{\bar{V}_{out}^2}{A_v^2} = 4kT\gamma \cdot \frac{1}{g_m}$$

2. 闪烁噪声: $\bar{V}_n^2 = \frac{k}{C_{ox} W L} \cdot \frac{1}{f}$, $\bar{I}_n^2 = \frac{k}{C_{ox} W L} \cdot \frac{1}{f} \cdot g_m^2$



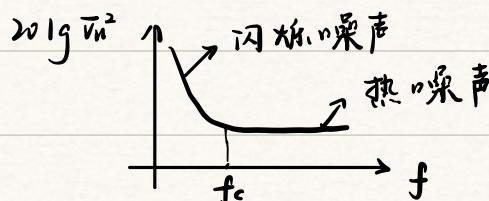
3. 转角频率:

热噪声 = 闪烁噪声时的频率

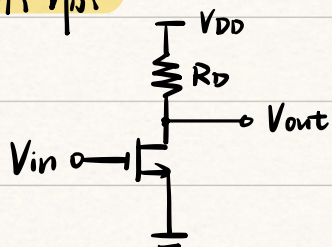
$$4kT\gamma = g_m^2 \cdot \frac{k}{C_{ox}WL} \cdot \frac{1}{f}$$

$$\Rightarrow f_c = \frac{k}{C_{ox}WL} \cdot \frac{g_m}{4\gamma kT}$$

低频时: 闪烁噪声是主要的
高频时: 主要是热噪声



共源



$$A_v = -g_m R_D$$

① 闪烁噪声

$$\overline{V_{out,f}}^2 = \overline{V_{in}}^2 \cdot A_v^2 = g_m^2 R_D^2 \cdot \frac{k}{C_{ox}WL} \cdot \frac{1}{f}$$

$\overline{V_{out}}^2$ 随 R_D 增大而增大 $\Rightarrow$ 一般用输入噪声衡量

② 热噪声

$$\overline{V_{out, TM}}^2 = \overline{I_n}^2 \cdot R_{out}^2 = 4kT\gamma g_m R_D^2$$

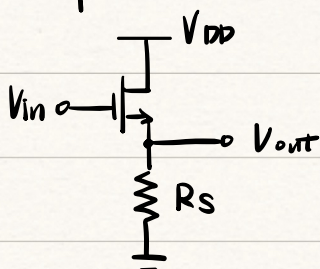
③ 电阻热噪声

$$\overline{V_{out, TR}}^2 = 4kT R_D$$

$$\overline{V_{out}}^2 = \sum_{i=1}^3 \overline{V_{out,i}}^2 = 4kT R_D + 4kT\gamma g_m R_D^2 + \frac{k}{C_{ox}WL} \cdot \frac{1}{f} \cdot g_m^2 R_D^2$$

$$\overline{V_{in}}^2 = \frac{\overline{V_{out}}^2}{A_v^2} = \frac{4kT}{g_m^2 R_D} + \frac{4kT\gamma}{g_m} + \frac{k}{C_{ox}WL} \cdot \frac{1}{f}$$

共漏



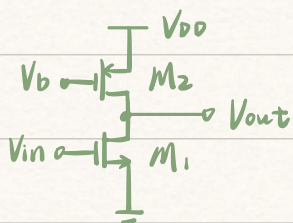
$$A_v = \frac{g_m R_s}{1 + g_m R_s}$$

$$\overline{V_{in, TM}}^2 = 4kT\gamma \cdot \frac{1}{g_m}$$

$$\overline{V_{out, TR}}^2 = 4kT \cdot \frac{1}{R_s} \cdot (R_s \parallel \frac{1}{g_m})^2$$

$$\overline{V_{in, T}}^2 = 4kT\gamma \cdot \frac{1}{g_m} + 4kT \cdot \frac{1}{g_m^2 R_s}$$

eg. 只考虑 Mos 热噪声, 考虑 r_o , 求 $\overline{V_{in}^2}$



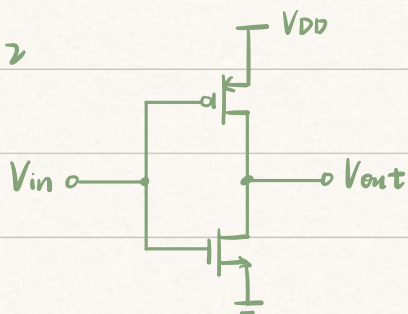
$$A_v = -g_{m1}(r_{o1} \parallel r_{o2})$$

$$R_{out} = r_{o1} \parallel r_{o2}$$

$$\overline{V_{out}^2} = 4kTV(g_{m1} + g_{m2})(r_{o1} \parallel r_{o2})^2$$

$$\overline{V_{in}^2} = \frac{\overline{V_{out}^2}}{A_v^2} = 4kTV \cdot \frac{g_{m1} + g_{m2}}{g_{m1}^2}$$

eg. 2



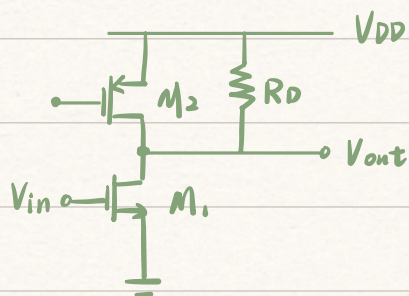
$$A_v = (g_{m1} + g_{m2})(r_{o1} \parallel r_{o2})$$

$$\overline{I_{out}^2} = 4kTV(g_{m1} + g_{m2})$$

$$\overline{V_{out}^2} = \overline{I_{out}^2} \cdot R_{out}^2 = 4kTV(g_{m1} + g_{m2})(r_{o1} \parallel r_{o2})^2$$

$$\overline{V_{in}^2} = \frac{\overline{V_{out}^2}}{A_v^2} = \frac{4kTV}{g_{m1} + g_{m2}}$$

eg. 3 求所有噪声 (一般 r_o 可以忽视时就不考虑)



$$\textcircled{1} R: \overline{V_{out,TR}^2} = 4kTR_D$$

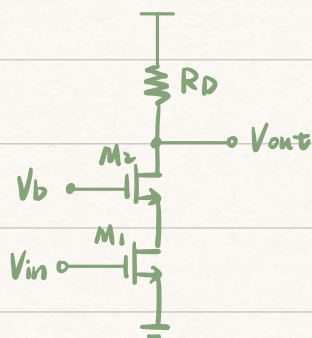
$$\textcircled{2} Mos: R_{out} = R_D$$

$$\overline{V_{out,IM}^2} = 4kTV(g_{m1} + g_{m2})R_D^2$$

$$\textcircled{3} Mos: \overline{V_{out,fn}^2} = \frac{K_N}{C_{ox}(WL)_1} \cdot \frac{1}{f} \cdot g_{m1}^2 R_D^2 + \frac{K_P}{C_{ox}(WL)_2} \cdot \frac{1}{f} \cdot g_{m2}^2 R_D^2$$

$$\overline{V_{in}^2} = \frac{\sum \overline{V_{out}^2}}{A_v^2}, \quad A_v = -g_m R_D$$

eg.4 求输入热噪声



$$\textcircled{1} \overline{V_{out,TR}}^2 = 4kTR_D$$

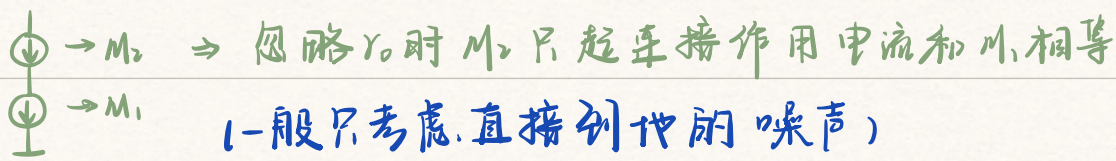
$$\textcircled{2} \overline{I_{n,TM1}}^2 = 4kT\gamma g_{m1}$$

$$\overline{V_{out,TM1}}^2 = 4kT\gamma g_{m1}R_D^2$$

$$A_V = -g_{m1}R_D$$

$$\overline{V_{in}}^2 = \frac{\sum \overline{V_{out}}^2}{A_V^2} = 4kT \frac{1}{g_{m1}^2 R_D} + 4kT\gamma \cdot \frac{1}{g_{m1}^2}$$

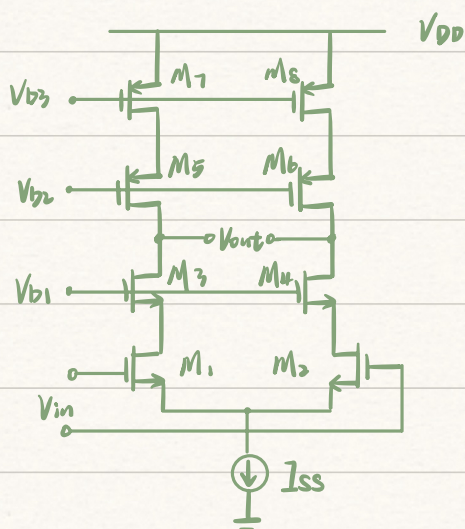
M_2 管热噪声对输入端影响很小，可忽略不计



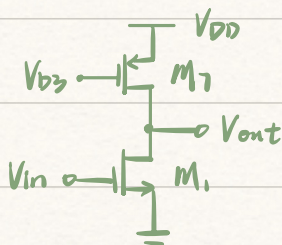
差分对

差动噪声 = 单端噪声 $\times 2$

eg. 求 $\overline{V_{in}}^2$



$$A_V = g_{m1} R_{out}$$



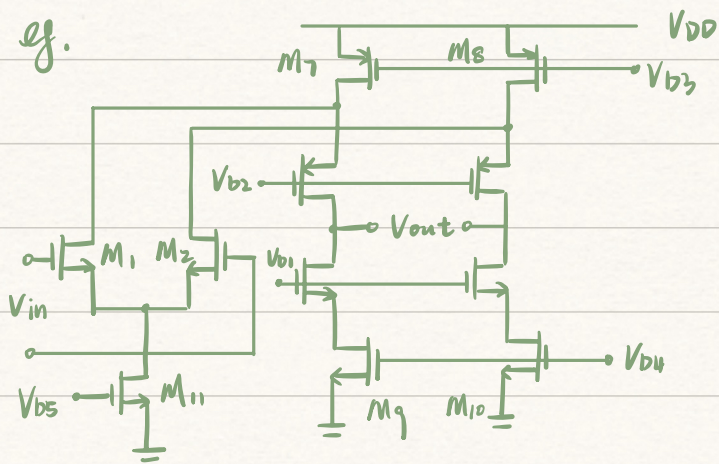
$$\overline{V_{out,H}}^2 = 4kT\gamma (g_{m1} + g_{m7}) R_{out}^2$$

$$\overline{V_{out,f}}^2 = \frac{K_N}{C_{ox}(WL)_1} \cdot \frac{1}{f} \cdot g_{m1}^2 R_{out}^2 + \frac{K_P}{C_{ox}(WL)_7} \cdot \frac{1}{f} \cdot g_{m7}^2 R_{out}^2$$

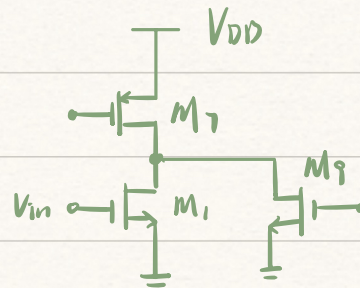
$$\overline{V_{in}}^2 = 2 \cdot \frac{\overline{V_{out}}^2}{A_V^2} = 8kT\gamma \left(\frac{1}{g_{m1}^2} + \frac{g_{m7}}{g_{m1}^2} \right) + \frac{2K_N}{C_{ox}(WL)_1} \cdot \frac{1}{f} + \frac{2K_P}{C_{ox}(WL)_7} \cdot \frac{1}{f} \cdot \frac{g_{m7}^2}{g_{m1}^2}$$

$\overline{V_{in}}^2$ 和 R_{out} 无关，可以不计算 R_{out}

eg.



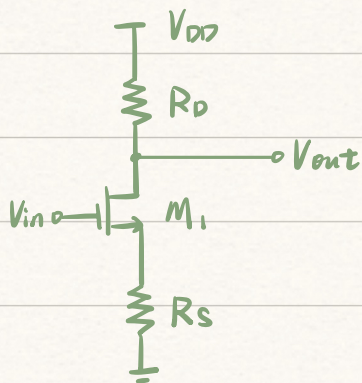
$$A_V = -g_{m1} R_{out}$$



$$\overline{V_{out,T}}^2 = 4kT\gamma (g_{m1} + g_{m7} + g_{m9}) R_{out}^2$$

$$\overline{V_{out,f}}^2 = \frac{k_P}{C_{ox}(WL)_T} \cdot \frac{1}{f} \cdot g_{m1}^2 R_{out}^2 + \frac{k_N}{C_{ox}(WL)_1} \cdot \frac{1}{f} \cdot g_{m1}^2 R_{out}^2 + \frac{k_N}{C_{ox}(WL)_9} \cdot \frac{1}{f} \cdot g_{m9}^2 R_{out}^2$$

eg. 求 $\overline{V_{in,T}}^2$

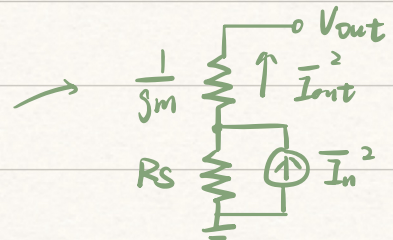


$$R_{out} = R_D \quad A_V = \frac{g_m R_D}{1 + g_m R_S}$$

$$\overline{V_{out,TD}}^2 = 4kT R_D$$

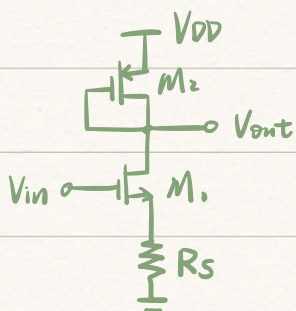
$$\overline{V_{out,TS}}^2 = \frac{4kT}{R_S} \cdot \left(\frac{R_S}{\frac{1}{g_m} + R_S} \right)^2 \cdot R_D^2$$

$$\overline{V_{in,TM}}^2 = 4kT\gamma \cdot \frac{1}{g_m}$$



$$\overline{V_{in}}^2 = \frac{\overline{V_{out,TD}}^2}{A_V^2} + \overline{V_{in,TM}}^2 = 4kT R_D \left(\frac{1 + g_m R_S}{g_m R_D} \right)^2 + 4kT R_S + 4kT\gamma \cdot \frac{1}{g_m}$$

eg. 求 $\overline{V_{in,T}}^2$



$$A_V = \frac{g_{m1}}{1 + g_{m1} R_S} \cdot \frac{1}{g_{m2}} \cdot \overline{I_{out}}^2 \cdot R_{out}^2$$

$$\overline{V_{out,TP}}^2 = \frac{4kT}{R_S} \cdot \left(\frac{R_S}{\frac{1}{g_m} + R_S} \right)^2 \cdot \frac{1}{g_{m2}^2}$$

$$\overline{V_{out,T2}}^2 = 4kT\gamma g_{m2} \cdot \left(\frac{1}{g_{m2}} \right)^2$$

$$\overline{V_{in,T1}}^2 = 4kT\gamma \cdot \frac{1}{g_{m1}}$$

$$\overline{V_{in}}^2 = 4kT\gamma \cdot \frac{1}{g_{m1}} + 4kT\gamma \cdot g_{m2} \cdot \left(\frac{1 + g_{m1} R_S}{g_{m1}} \right)^2 + 4kT R_S$$

5. 摆幅

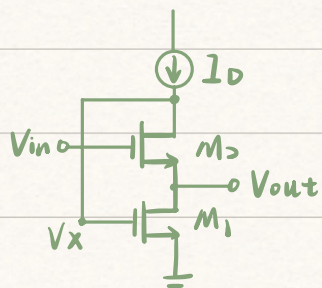
管子处于饱和区时, V_{out} 变化范围

① 一般题里会给 I_D . 由 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \Rightarrow V_{GT}$

② $V_{DS} \geq V_{GT}$

对于双端输出差分对: 摆幅 = 单边摆幅 $\times 2$

eg.



$C_{ox} = 5 \times 10^{-5} F$, $I_D = 60 \mu A$, $V_{TH} = 0.5 V$, $\mu_n = 300 cm^2/(V \cdot s)$

$\frac{W}{L} = 5$, 求 V_{out} 范围

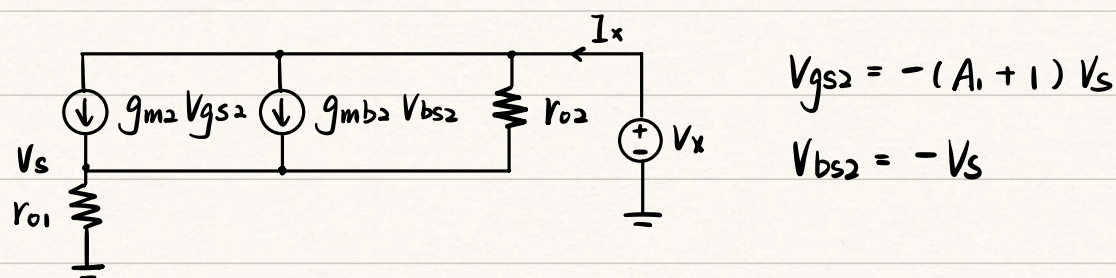
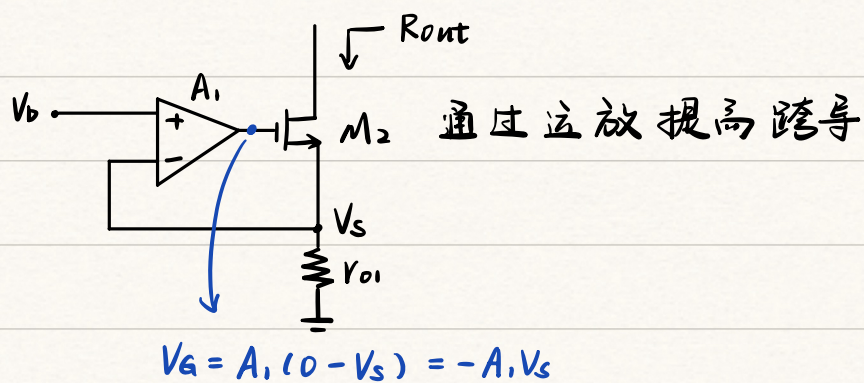
① $I_{D1} = I_{D2} = 60 \mu A \Rightarrow V_{GT1} = V_{GT2} = 0.4 V$

② $M_1 \begin{cases} V_{GT} = V_x - V_{TH} = 0.4 V \Rightarrow V_x = 0.9 V \\ V_{DS} = V_{out} \geq V_{GT} = 0.4 V \end{cases}$

M_2 : $V_{DS} = V_x - V_{out} \geq V_{GT} = 0.4 V \Rightarrow V_{out} \leq 0.5 V$

综上: $0.4 V \leq V_{out} \leq 0.5 V$

增益提高技术



$$R_{out} = \frac{V_x}{I_x} = r_{O1} + r_{O2} + (A_1 + 1) g_{m2} r_{O1} r_{O2} + g_{mb2} r_{O1} r_{O2}$$

$$= r_{O2} + [1 + [(A_1 + 1) g_{m2} + g_{mb2}] r_{O2}] r_{O1}$$

未提高时: $R_{out} = r_{O2} + [1 + (g_{m2} + g_{mb2}) r_{O2}] r_{O1}$

相当于 M_2 管的跨导增大至 $(1 + A_v) g_{m2}$

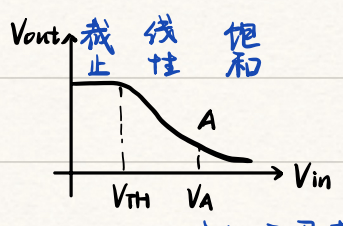
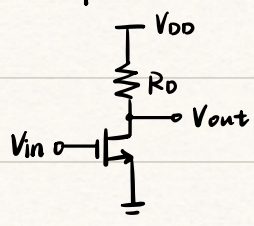
草图

共漏：截止 → 饱和

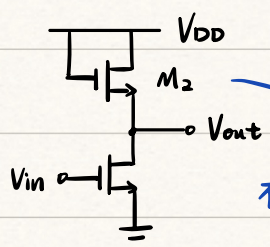
1. 放大管 截止区 → 饱和区 → 线性区，负载管：截止区 → 线性区 → 饱和区

共源：

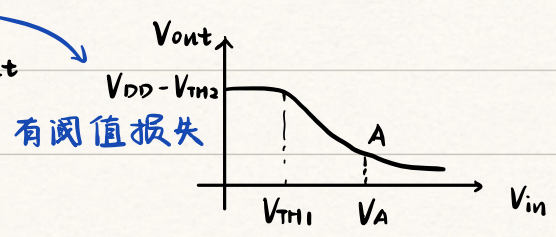
截止： $V_{GS} = V_{in} \leq V_{TH}$ ， $V_{out} = V_{DD}$



不用求具体数值，标出有转折点即可



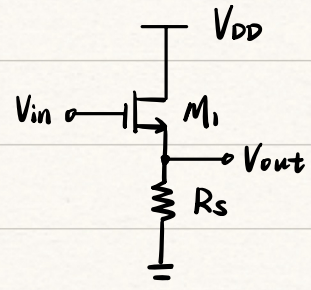
截止： $V_{GS} = V_{in} \leq V_{TH1}$ ， M_2 也截止， $V_{GS2} = V_{TH2} \Rightarrow V_{out} = V_{DD} - V_{TH2}$



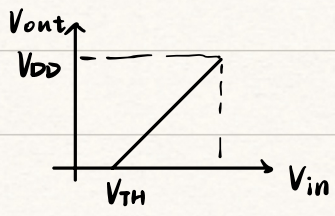
负载管要满足 $V_{GS} \geq V_{TH}$

共漏：

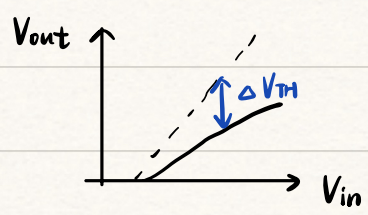
截止： $V_{GS} = V_{in} \leq V_{TH} \Rightarrow V_{out} = 0$



不考虑体效应：
(源跟随器)

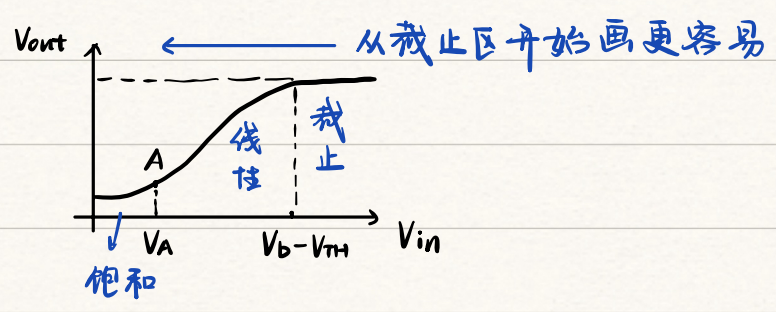
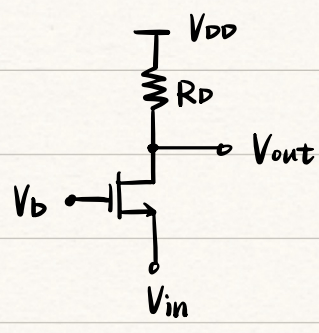


考虑体效应：
($V_{TH} \uparrow$)



共栅：

截止： $V_{GS} = V_b - V_{in} \leq V_{TH} \Rightarrow V_{out} = V_{DD}$



从截止区开始画更容易

2. 差分对

注意双端输出摆幅加倍

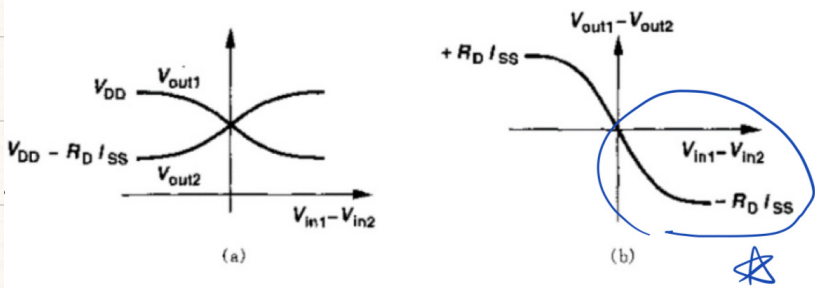


图 4.7 差分对的输入-输出特性

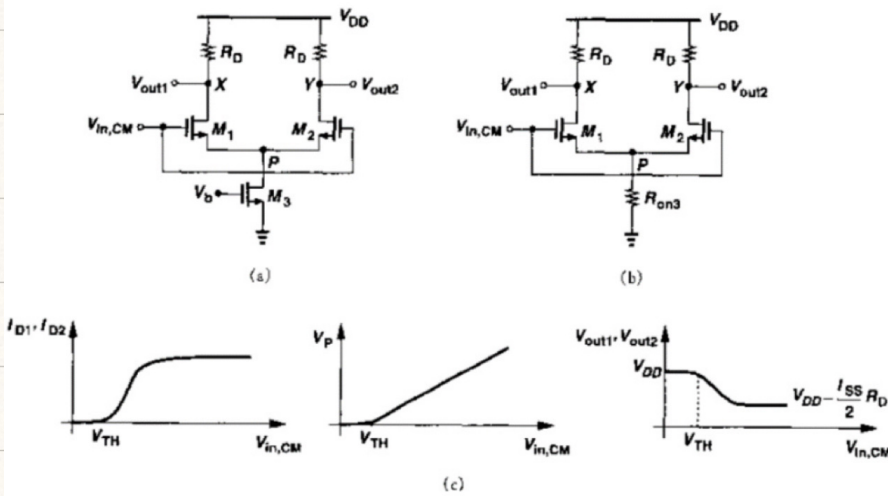
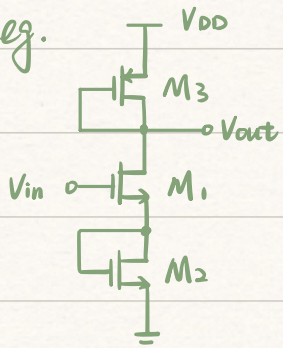
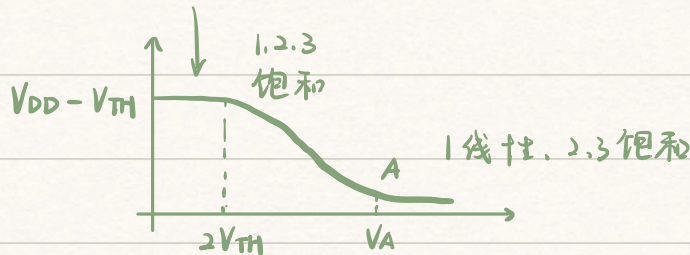


图 4.8 (a) 检测输入共模电压变化的差分对电路; (b) M_3 管工作在深线性区时的等效电路; (c) 共模输入-输出特性曲线

eg. $V_{in} : 0 \rightarrow V_{DD}$. 画 V_{out} 随 V_{in} 变化的草图



截止: $M_1: V_{GS} = V_{in} - V_{TH(2)} \geq V_{TH} \Rightarrow V_{in} \geq 2V_{TH}$



对于二极管接法

始终有 $V_{DS} = V_{GS} > V_{GS} - V_{TH}$

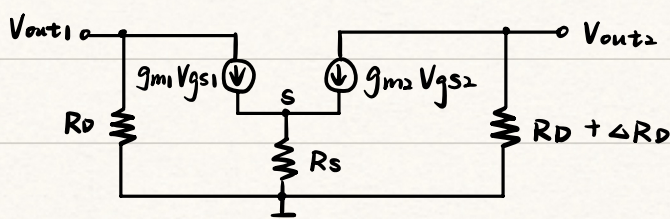
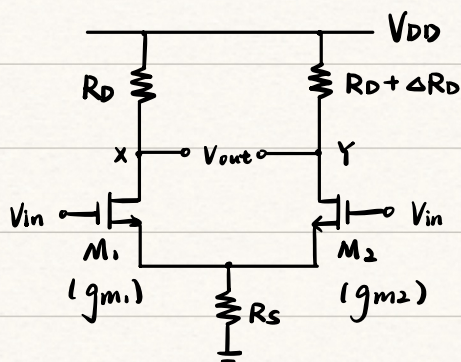
$\Rightarrow$ 不存在线性区

过程: 截止 $\rightarrow$ 饱和

差分对不匹配

- 影响 $\left\{ \begin{array}{l} \text{不影响差模增益 (近似)} \\ \text{共模增益不为0} \end{array} \right.$

2. 求法：画小信号图



$$\begin{cases} V_{gs1} = V_{in} - V_s, \quad V_{gs2} = V_{in} - V_s \\ V_s = (g_{m1} V_{gs1} + g_{m2} V_{gs2}) \cdot R_s \end{cases}$$

$$A_c = \frac{V_{out1} - V_{out2}}{V_{in}} = \frac{g_{m2} V_{gs2} (R_D + \Delta R_D) - g_{m1} V_{gs1} R_D}{V_{in}} = \frac{(g_{m2} - g_{m1}) R_D - g_{m2} \Delta R_D}{1 + (g_{m1} + g_{m2}) R_s}$$

$$\left\{ \begin{array}{l} \text{跨导不匹配: } A_c = \frac{(g_{m2} - g_{m1}) R_D}{1 + R_{ss} (g_{m1} + g_{m2})} \end{array} \right.$$

$$\left\{ \begin{array}{l} \text{电阻不匹配: } A_c = \frac{-g_{m2} \Delta R_D}{1 + 2g_m R_s} \end{array} \right.$$

3. 共模抑制比 $CMRR = \left| \frac{A_d}{A_c} \right|$

运放设计

先考虑大信号参数：功率、摆幅

分配过驱动电压原则： $\mu_{P1}V_{odp1} = \mu_{n1}V_{odn1}$

再考虑小信号参数：增益、带宽